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Emerging Per- and Polyfluoroalkyl Substances in Tap Water from the American Healthy Homes Survey II

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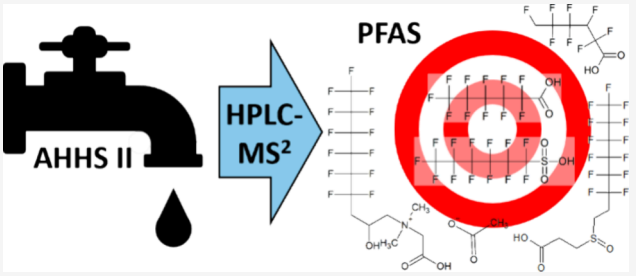
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ABSTRACT: Humans experience widespread exposure to anthropogenic per- and polyfluoroalkyl substances (PFAS) through various media, which can lead to a wide range of negative health impacts. Tap water is an important source of exposure in communities with any degree of contamination but routine or large-scale PFAS monitoring often depends on targeted analytical methods limited to measuring specific PFAS. We analyzed 680 tap water samples from the American Healthy Homes Survey II for PFAS using non-targeted analysis (NTA) to expand the range of detectable PFAS. Based on detection frequency and relative abundance, about half of the identified PFAS were found only by NTA. We identified (with varying degrees of confidence) 75 distinct PFAS, including 57 exclusively detected by NTA. The identified PFAS are members of seven structural subclasses differentiated by their head groups and degree of fluorination. Clustering analysis categorized the PFAS into four coabundance groups dominated by specific PFAS subclasses. One group uniquely identified by NTA contains zwitterionic PFAS and other PFAS transformation products which are likely associated with aqueous firefighting foam contaminants in a small number of spatially correlated samples. These results help further characterize the scope of exposure to emerging PFAS experienced by the U.S. population via tap water and augment nationwide targeted-PFAS monitoring programs.



KEYWORDS: *non-targeted analysis, drinking water, aqueous firefighting foam, mass spectrometry, exposure*

1. INTRODUCTION

Per- and polyfluoroalkyl substances (PFAS) are a class of anthropogenic chemicals with widespread evidence of human exposure; National Health and Nutrition Examination Survey (NHANES) data indicate that most people in the United States have measurable levels of PFAS in their blood serum.[1] Evaluating PFAS exposures is important because some PFAS are associated with a wide range of potential health effects.[2−5] The most well-studied PFAS have half-lives in the human body on the order of years,[6] meaning body burden can accumulate over long exposure. Exposure sources for PFAS have been extensively reviewed and include contaminated food, drinking water, air, and indoor media including consumer products.[2,7−10] Ingestion of drinking water is an important pathway for exposure to PFAS, especially in contaminated communities.[9,11−14]

Concerns about persistence, exposure, and impacts to human health and the environment led to a voluntary phaseout of the use of select PFAS by some manufacturers in the early mid 2000s.[15−17] The phaseout was successful at reducing exposure to perfluorooctanesulfonate (PFOS) and perfluorooctanoic acid (PFOA), as well as other long-chain perfluoroalkyl carboxylic (PFCAs) and sulfonic acids (PFSAs) which are now termed "legacy" PFAS.[1,18,19] However, this drove a shift to lower molecular weight analogues, such as perfluorobutanesulfonic acid (PFBS), and structural analogues with different functional groups such as ether linkages.[20−22] Targeted PFAS panels have been expanded to monitor for the PFAS used in the wake of the phaseout. For example, targeted approaches under the Fifth Unregulated Contaminant Monitoring Rule (UCMR5) are being used to monitor for twenty-nine PFAS in drinking water supplies throughout the United States.[23] However, thousands of "emerging" PFAS with explicit chemical structures have been

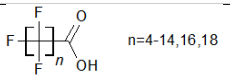
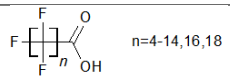
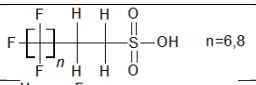
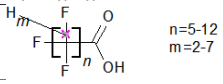
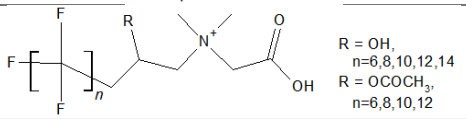
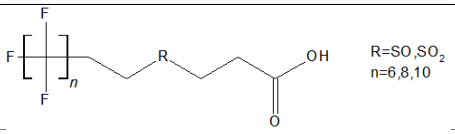
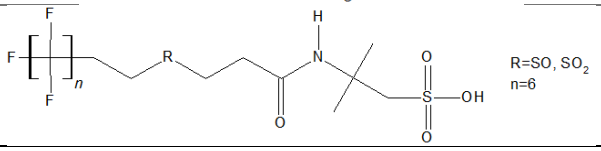
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Table 1. PFAS Subclasses Detected in AHHS II Tap Water Samples

Sub-class	Markush Structure	Abbrev.	n	Confidence	Detect Frequency[a] (n=680)
Perfluoroalkyl carboxylic acids		PFCA	10	1	1-229
Perfluoroalkyl sulfonic acids		PFSA	9	1-3	3-489
Fluorotelomer sulfonates (n:2)		n:2 FTS	2	2-3	15-23
H-substituted PFCAs		Hn PFCA	21	3-5	2-261
Fluorotelomer substituted betaines (n:3)		n:3 FTRB	11	2-3	11-93
Fluorotelomer sulfinyl/sulfonyl propanoic acids (n:2)		n:2 FTRPrA	6	2-3	3-72
Fluorotelomer sulfinyl/sulfonyl amido sulfonates (n:2)		n:2 FTRAmS	2	3	37-62
Other			14	2-5	5-248

[a]The range for Detect Frequency indicates the minimum and maximum number of samples in which a member of the PFAS subclass was detected. Individual detection frequencies for each compound are given in Table S4.

reported.[21,24,25] The extent to which these PFAS, invisible to targeted methods, contaminate media relevant to human exposure such as drinking water is currently unknown.

The presence of potentially thousands of unmonitored emerging PFAS and their increase in proportion over time[14] makes additional methods of PFAS screening and identification beyond targeted analysis necessary to capture the full breadth of PFAS in exposure sources such as drinking water. Suspect screening and non-targeted analysis (NTA) approaches have rapidly become important tools to identify emerging PFAS and other emerging contaminants in drinking water and associated raw surface and groundwater sources,[26−33] as well as in biological media of populations known to be exposed via their drinking water.[26,34,35] For instance, NTA was used to identify HFPO−DA (a.k.a. GenX) at drinking water intakes in the Cape Fear River watershed of North Carolina[36] with later standard-based quantification.[37] In this way, NTA can be used to screen for emerging contaminants for which little exposure[38] or toxicity data[39] are generally available. For instance, a network of European researchers has developed guidance on the application of NTA to contaminant screening motivated by the broader detection space enabled by NTA, technological improvements, and interest in NTA from regulators.[40]

The American Healthy Homes Survey (AHHS) was a national sampling effort carried out as a joint effort between the U.S Department of Housing and Urban Development (HUD) and U.S. Environmental Protection Agency (EPA) in both 2005 (AHHS I) and 2017 (AHHS II). The AHHS studies were primarily designed to evaluate hazards such as lead, pesticides, and allergens in United States homes where children may live[41] by collecting soil, dust, paint, and tap (drinking) water as well as demographic information. The most recent analyses of the collected samples have included metals other than lead[42] and targeted analyses of legacy PFAS on drinking water samples, paired with household dust.[43] In this study we report a follow-on analysis of PFAS in AHHS II drinking water samples by NTA, to better characterize the diversity and distribution of emerging PFAS to which United States residents are exposed through their drinking water. We describe the occurrence patterns in tap water of several subclasses of PFAS that are not analyzed by most targeted methods.

2. METHODS

Expanded methods are discussed in Section S1.

2.1. Sample Collection. The AHHS II study recruitment and sample collection procedures are described by Bradham et al.[42] and U.S. HUD.[41] Although the sampling was optimized with respect to the prevalence of lead, the sample is still representative of PFAS incidence on a national basis; as such, it can be considered a nationally representative sample of all permanently occupied, noninstitutional housing where children may live. Residents collected cold tap water in wide-mouth HDPE bottles using a single-day composite sampling approach. After shipment to the laboratory, samples were acidified to 2% (w/w) nitric acid and held at 4 °C.

2.2. Analytical Methods. Samples were processed as for targeted analysis,[43] using a modification of EPA Method 533. Single tap water samples from each home (n = 680) were weighed and pooled with 10 mL methanolic rinsate of the original sample container. Samples were spiked to 50 ng/L with

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isotope-labeled PFAS internal standard mixture (Wellington MPFAC-C-ES) prior to extraction. A 250 mL aliquot was loaded onto a pre-equilibrated weak anion exchange (WAX) solid-phase extraction (SPE) cartridge (Waters Oasis) using positive-displacement pumping. Cartridges were then rinsed with 5 mL pH 4 buffer followed by 5 mL methanol. Cartridges were eluted to 15 mL conical tubes using 0.1% ammonium hydroxide in methanol. Eluent was held at 40 °C and evaporated to <500 μL using dry N_2 gas at 12 psi, then reconstituted to 500 μL. Aliquots of sample extracts were combined 1:3 v:v with aqueous mobile phase in HPLC autosampler vials.

Sample extracts were analyzed via UPLC-MS/MS on a Thermo Orbitrap Fusion tribrid mass spectrometer. Separation was achieved using reverse-phase liquid chromatography and a Waters Acquity C18 column including a delay column. Electrospray ionization (ESI) in negative polarity was applied, with positive mode applied to selected samples. MS^2 and MS^3 fragmentation was applied following a preferred-ion data-dependent acquisition (DDA) procedure. Chromatography and mass spectrometry parameters are given in Tables S1 and S2.

Acidified standards and method blanks were prepared. Continuous calibration standards were analyzed every ten samples. One method blank was prepared and analyzed for each of 19 sample preparation batches.

2.3. Data Processing. NTA was performed in Thermo Compound Discoverer 3.3.2.31 (CD) including custom scripts in R for feature prioritization. Feature masses were matched to the EPA's Distributed Structure Searchable Toxicity (DSSTox) database of molecular masses and formulas[44] and a database of molecular masses and formulas of PFAS associated with aqueous film-forming foam (AFFF) and other sources.[45] MS^2 spectra were matched against spectra from Mass Bank of North America (MoNA), December 2019[46] and mzCloud version July 2021.[47] Prioritization also used maximum ion abundance, fluorine-containing predicted formulas, negative mass defect, potential homologous series, and common PFAS MS^2 fragments and neutral losses. CD analysis parameters can be found in Section S1.3 and Table S3. All prioritized features were scored on the PFAS-specific Schymanski scale;[48,49] justifications for each rating are provided in Section S2.

Identified molecular features were postprocessed to adjust for isotope-labeled internal standard response and filter non-reproducible features or those originating from adducts and in-source fragments. The population of method blanks was used to construct an effective detection limit at: (method blank median) + 10x (method blank standard deviation).

Statistical and geospatial analyses were performed on all PFAS identified in at least one sample. These analyses include per-sample PFAS detections and abundances (Section 3.2.1), geospatial distributions of total PFAS abundance (Section 3.2.2), coabundance patterns among PFAS (Section 3.3), and PFAS profile similarity between samples (Section 3.4). Co-abundance patterns were evaluated using individual quadratic assignment procedure (QAP) methods.[50]

3. RESULTS AND DISCUSSION

3.1. Detected PFAS. *3.1.1. Detected PFAS Classification.* PFAS identified via NTA were categorized into chemical subclasses and are described with a generalized Markush structure in Table 1, along with detection frequencies, assigned confidence scores, and maximum abundances. Justification for each detected PFAS identification (Section S2) including MS^2 spectra (Figures S1−S8) and metadata for individual PFAS (Tables S4, S6) is provided in the Supporting Information. Below, we provide a discussion of the most salient detection and identification information, followed by a discussion of past environmental detections and/or patent use information.

The perfluoroalkyl carboxylic (PFCA) and sulfonic acids (PFSA), i.e., the "legacy" PFAS, were mostly assigned by direct comparison to standards (confidence 1) and are frequently included in targeted methods, e.g., EPA method 533.[51] The n:2 fluorotelomer sulfonates (FTS) are also frequently included in targeted methods, but were either detected by spectral match to library spectra (confidence 2a, 6:2 FTS) or by a RT match with limited MS^n information (3b, 8:2 FTS) in this study. Spectra of both FTS contain the expected $[M\text{-}H\text{-}HF]^-$ fragment, and the former also contains the expected SO_3^- and HSO_3^- fragments.

The hydrogen-substituted (H-substituted) PFCAs detected contain between two and seven H substitutions for fluorine (F). This subclass likely includes fluorotelomer carboxylic acids (FTCA) with even numbers of H substitutions as well as isomers involving branching and/or H-substitution at different carbon (C) atoms along the chain. A high degree of isomerization is indicated by broad chromatographic peaks which in some instances contain multiple discrete peaks (Figure S9). This subclass also includes compounds with an odd number of H substitutions. MS^2 spectra of H-substituted PFCAs made at confidence 3 (the best possible given the range of likely isomers) all contain at least one decarboxylation fragment with accompanying HF loss, i.e., $[M\text{-}H\text{-}CO_2\text{-}nHF]^-$ (Table S6).

The n:3 substituted fluorotelomer betaines are zwitterionic species which were principally detected as adducts in negative polarity. Several members of the subclass were detected as $[M + H]^+$ peaks in positive polarity, and their structures were confirmed using their positive polarity MS^2 spectra (Figure S8). The fluorotelomer betaine MS^2 spectra exhibited the $C_4H_{10}NO_2^+$ (m/z 104.071) fragment in ESI(+) mode in common with five subclasses of perfluoroalkyl (i.e., non-fluorotelomer) sulfonamido betaines detected in AFFF, PFAS formulations for consumer products, and/or AFFF-impacted groundwater.[27] Fragments of partially fluorinated tails are also visible in ESI(+) mode. The spectra also exhibited $C_4H_8NO_2^-$ and $C_6H_{10}NO_3^-$ (m/z 102.056 and 144.066 respectively) fragments in ESI(−) mode, as well as a series of HF-loss fragments derived from the perfluorinated tail and adjoining nonfluorinated telomer (Figure S4). In ESI(−) mode, all betaines were primarily detected as adducts with acetate or nitrate owing to the permanent positive charge on the betaine quaternary N.

The n:2 fluorotelomer sulfinyl and sulfonyl propanoic acids readily ionize for detection in ESI(−) mode. Their MS^2 spectra match spectra reported in microcosms or environmental samples.[27,52] They exhibit two parallel sets of HF-loss tail fragments as well as SO_2F^- (only sulfonyls) and $HSO_xC_2H_4COO^-$ fragments derived from the headgroup (Figure S2). The fluorotelomer sulfinyl/sulfonyl amido sulfonates were detected only with a confidence of 3d, due to comparatively low abundances (never exceeding 1E6) and a consequent lack of DDA MS^2 data.

In addition, 14 other potential PFAS are identified that do not fall under these seven subclasses. These include polyfluorinated ethers Nafion byproduct 2 (confidence 2b) and fluoro-(heptafluoropropoxy)acetic acid (3a), unsaturated PFOS (3a), 8:3 fluorotelomer 2-hydroxy trimethylammonium (3c), and

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bistriflimide (4). The nine confidence-5 detections often have no available MS^2 data, but match masses of known PFAS.

Aside from the PFAS detected in this data set, there were several detections of non-PFAS aromatic compounds at confidence 2a and were likely affected by the acidification with 2% nitric acid (Section S2, Table S5).

3.1.2. Prior Detection and Use Information. The H-substituted PFCAs likely consist in part of FTCAs with many industrial and commercial uses as surface coatings. There are some instances of targeted detection of FTCAs in environmental samples. For instance, 8:3 FTCA was targeted but not detected in surface water,[53] rainwater,[54] and tap water,[55] but was detected in landfill gas condensate by targeted analysis,[56] marine sediments by targeted analysis,[57] and beluga whale liver by suspect screening analysis.[58] EPA method 1633 includes 3:3, 5:3, and 7:3 FTCA as analytes.[59] The FTCAs have even numbers of H substitutions. The H-substituted PFCAs also contains members with an odd number of H substitutions, which have fewer reported industrial uses, e.g. fireproofing[60] but could be transformation products of other PFAS. For instance, microbially mediated reductive dechlorination of chlorinated PFAS is known to produce H-substituted PFAS,[61] although chlorinated PFAS were not identified in AHHS II samples. However, H1-PFCAs, the most commonly reported odd-H-substituted PFCAs and which have been reported as environmental transformation products,[27,62−64] were not detected in this study.

Little information is available on the uses and prevalence of the betaine subclass, although they do appear in a mass list of AFFF-associated PFAS[45] and the acetoxy series appears in a patent for hydraulic fracturing fluids.[65] Five separate betaine homologous series were detected in several AFFF formulations, related consumer product PFAS formulations, and impacted groundwater.[27] The series detected in that study all contain a sulfonamido moiety, consist of members differing by CF_2 units, and likely consist of several branched isomers. Due to the acidic preservation used in this study, it is uncertain whether in-sample oxidation of previously studied PFAS formulations from Barzen-Hanson et al. is the origin or whether these species are a new addition to the environmentally relevant PFAS betaine subclass. It is unlikely that betaines formed by in-source rearrangement because possible parent compounds were not detected.

The n:2 fluorotelomer sulfinyl and sulfonyl propanoic acids have no reported industrial or commercial uses, to our knowledge, but the sulfonyl series was detected in groundwater impacted by AFFF[27] and both series are believed to be related to contents of AFFF by oxidation and hydrolysis of precursor amido sulfonates.[66] The fluorotelomer sulfinyl/sulfonyl amido sulfonates were detected only with a confidence of 3d, but were also identified by Houtz et al.[66] in wastewater treatment influent and effluent. Houtz identified these as likely oxidized precursors to n:2 fluorotelomer sulfinyl and sulfonyl propanoic acids. Both the 6:2 sulfonyl propanoic acid and amido sulfonate were detected in a survey of tap-collected drinking water in Québec, alongside other emerging PFAS attributed most likely to an AFFF source.[31]

3.2. PFAS Detections and Abundances. *3.2.1. Detection and Abundance Distributions.* Detection distributions are presented in Figure 1a. 80% of tested water samples contain at least one legacy PFAS. On average, AHHS II tap water samples contain detectable levels of 3.5 legacy PFAS. A recent national study of legacy PFAS in tap water detected PFAS in 30% of measured tap water samples, and estimated that quantifiable

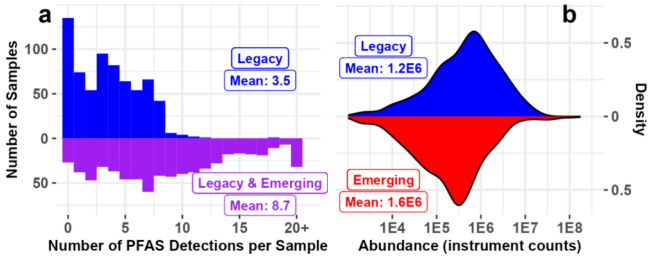


Figure 1. PFAS detected across n = 680 tap water samples. Legacy results (blue) come from 18 targeted PFCAs/PFSAs (C5, C6, C8, C10−C14, C16, C18 PFCAs; C4−C10, C12 PFSAs); emerging results (red) include an additional 57 features identified as likely PFAS. Combined legacy and emerging results (purple) include both. (a) Distribution and mean number of features identified in each sample (20+ IDs binned). (b) Distribution and mean total abundance (as chromatographic peak area) of PFAS features above blank level.

PFAS (with achievable detection limits, in reasonable sample sizes, using targeted analysis) likely exists in approximately half of all US drinking water samples.[67] The LODs in that study were somewhat higher, largely due to smaller sample volume depending on field sampling campaign (as small as 1 mL, vs 250 mL). Also, filter levels in this study, which are based on abundances in blanks, are less stringent than quantitation limits from targeted studies which rely on quantitation within error of standards. Therefore, these results are mutually consistent.

When emerging PFAS are included, 96% of all tap water samples contain at least one detectable PFAS, with an average of 8.7 PFAS at detectable levels in each sample. 4.7% of all tap water samples contain 20 or more individual PFAS. The abundance filter level used is conservative; 10x method blank abundance standard deviation should remove false positives from data processing or sample preparation (although not sampling).

Abundance distributions are presented in Figure 1b. Uncertainties associated with quantitative interpretation of NTA results make it challenging to compare abundances of PFAS measured using NTA with concentrations measured using targeted methods. Quantitative NTA is a developing field with this goal in mind[68,69] but a lack of knowledge of the ionization efficiencies and response factors of the emerging PFAS in this study make quantitative estimation difficult. This is especially true of the betaines, present often at high abundance, whose zwitterionic character and detection as adduct species differ markedly from legacy carboxylates and sulfonates. In Figure 1b, we sum the instrument abundances of all species directly. Individual chemical abundance distributions along with chemical-specific filter levels are shown in Figures S10−S17. An alternative classification based on whether compounds are targeted in EPA method 533[51] shows nearly identical detection and abundance distributions (Figure S18).

The mean abundances of legacy vs emerging PFAS indicate that on average, more than half the PFAS in these samples are likely from emerging PFAS. Assessment of wastewater treatment plant influent and effluent found that 21−88% of all PFAS falls outside the legacy PFAS,[70] consistent with this result. The legacy PFAS modal abundance is 2.5-fold higher than the emerging PFAS mode, but the mean emerging PFAS abundance is higher because it is dominated by a few samples with very high emerging PFAS abundances. This study suggests that current targeted methods may miss half or more of the PFAS present in

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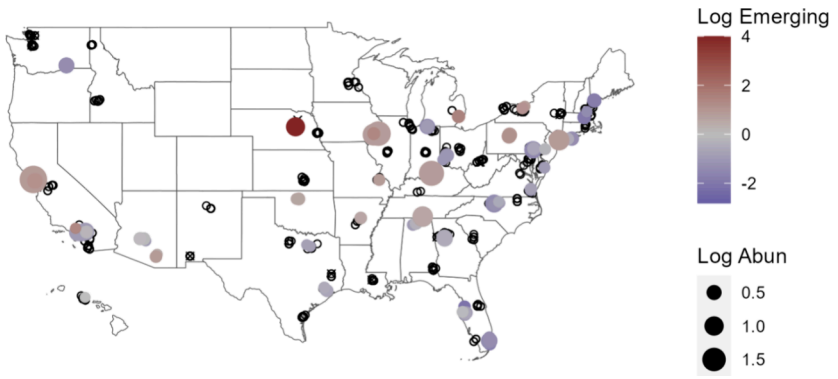


Figure 2. Locations of PFAS detections for $n = 653$ tap water samples in which at least one PFAS was detected (circles) and $n = 27$ tap water samples with no detected PFAS (X's). Color represents logarithmic deviation from expected emerging PFAS abundance (as chromatographic peak area; see text). Warmer (red) colors indicate a larger proportion of PFAS in a given sample is emerging PFAS. Circle size represents logarithmic deviation from mean total PFAS abundance. Colored circles ($n = 116$) represent locations with total PFAS abundance above the mean value. Empty circles ($n = 537$) represent locations with below-mean total PFAS abundance.

tap water, although any single emerging PFAS may exist in only a small subset of samples.

3.2.2. Geospatial Abundance Pattern. The geographic distribution of PFAS detections in this study are displayed in Figure 2. Circle size is controlled by logarithmic total PFAS abundance (Section S3). High abundances of PFAS are not geographically limited, although samples in the heavily urbanized northeastern US are more likely to exceed the mean abundance, likely due to local PFAS manufacturing and use. The mountain west had no sampling performed in AHHS II, thus the lack of PFAS detections can only be attributed to a lack of sampling This distribution broadly matches the trends observed by Smalling et al.[67] and DeLuca et al.[43] Alternative representations showing all samples >1 standard deviation above the mean (Figure S22) or all samples with any PFAS detected (Figure S26) are shown in the Supporting Information.

Circle color is controlled by logarithmic emerging PFAS abundance relative to what would be expected assuming legacy and emerging PFAS abundance were perfectly correlated; that is, color indicates where emerging PFAS are relatively larger/smaller contributors to total PFAS (Section S3). High emerging PFAS proportions are present across the country and not concentrated in the heavily urbanized northeastern US. Despite being present across the country, high proportions of emerging PFAS exhibit strong spatial autocorrelation, with a Global Moran's I[71] z-score of 17.15 and thus a probability of emerging from random chance of $<10^{-6}$. That is, sites with high ratios of emerging PFAS tend to cluster near each other, and sites with low ratios of emerging PFAS tend cluster near each other. This implies sources of emerging PFAS are not global, nor are they hyper-localized to single sites; instead, sources of emerging PFAS likely affect several nearby tap water locations at once.

3.3. Co-Abundance Patterns. *3.3.1. Co-Abundance Groups.* Correlations between detected PFAS were assessed using quadratic assignment procedure (QAP) to identify groups of PFAS which tend to exhibit similar abundance patterns in tap water. By doing so, individual PFAS might be identified that could serve as exposure proxies for others, and potential sources of PFAS mixtures contaminating tap water could be ascertained. A k-medoids cluster plot displaying coabundance clustering is shown in Figure 3. A corresponding heatmap of t- is shown in Figure S21. Four compound groups emerge from this analysis, corresponding to a high average silhouette width while maximizing chemical interpretability. Each group is dominated by one or two PFAS subclasses (Table 1), with a smaller number of one-off PFAS, structural outliers, and lower confidence compounds. Note that no structural information is included in the coabundance analysis; structural patterns are reflective of feature behavior and observed chemical similarities within clusters are due to the similar use/sources. Similar groupings emerge from Spearman rank-order correlation performed without Monte Carlo simulation (Figure S19) but the strength of QAP is the ability to test more completely for spurious correlations.

Compound group 1 consists largely of fluorotelomer betaine and fluorotelomer sulfinyl/sulfonyl propanoic acids, the majority of which are identified at confidence level 2b. 6:2 and 8:2 fluorotelomer sulfonates (3b), 8:3 fluorotelomer 2-hydroxy trimethylammonium (3c), and 6:2 fluorotelomer sulfinyl and sulfonyl propanoamido-dimethylethyl sulfonates (3d) are also in this compound group, all fluorotelomers whose structures have substantial similarity to other members. Another member of this group (6:3 fluorotelomer 2-hydroxy dimethylammonio carboxylic acid, 465.0612@4.846) may have a similar structure but MS^1 and MS^2 evidence for its structure are mixed (Section S2).

Additionally, compound group 1 contains C10 and longer PFCAs (except for PFTeDA) It also contains Nafion byproduct 2 (confidence 2b), a contaminant associated with ionic fluoropolymers identified in North Carolina[36,72] but detected in samples at several locations across the country in this data set. Unsaturated PFOS (3a) also groups with this group instead of the saturated sulfonates.

Compound group 2 consists largely of a subset H-substituted perfluorinated carboxylic acids at either confidence 3a (because the position of H-substitution and potentially branching is undetermined) or confidence 5b (because of a lack of MS^2 spectra). The only other compounds are confidence 5 matches.

Compound group 3 likewise consists of H-substituted perfluorinated carboxylic acids and confidence-5 compounds, but it also contains the C8−C12 PFSAs (confidence 1) and fluoro(heptafluoropropoxy)acetic acid (3a). Inverse relationships with compound group 4 are largely absent for members of this group, whereas they present for several members of compound group 2, which may drive the separation between compound groups 2 and 3. The groups are otherwise close in k-medoids clustering behavior (Figure S19).

Compound group 4 contains the remaining short-chain PFCAs and PFSAs, as well as PFTeDA. The group also contains

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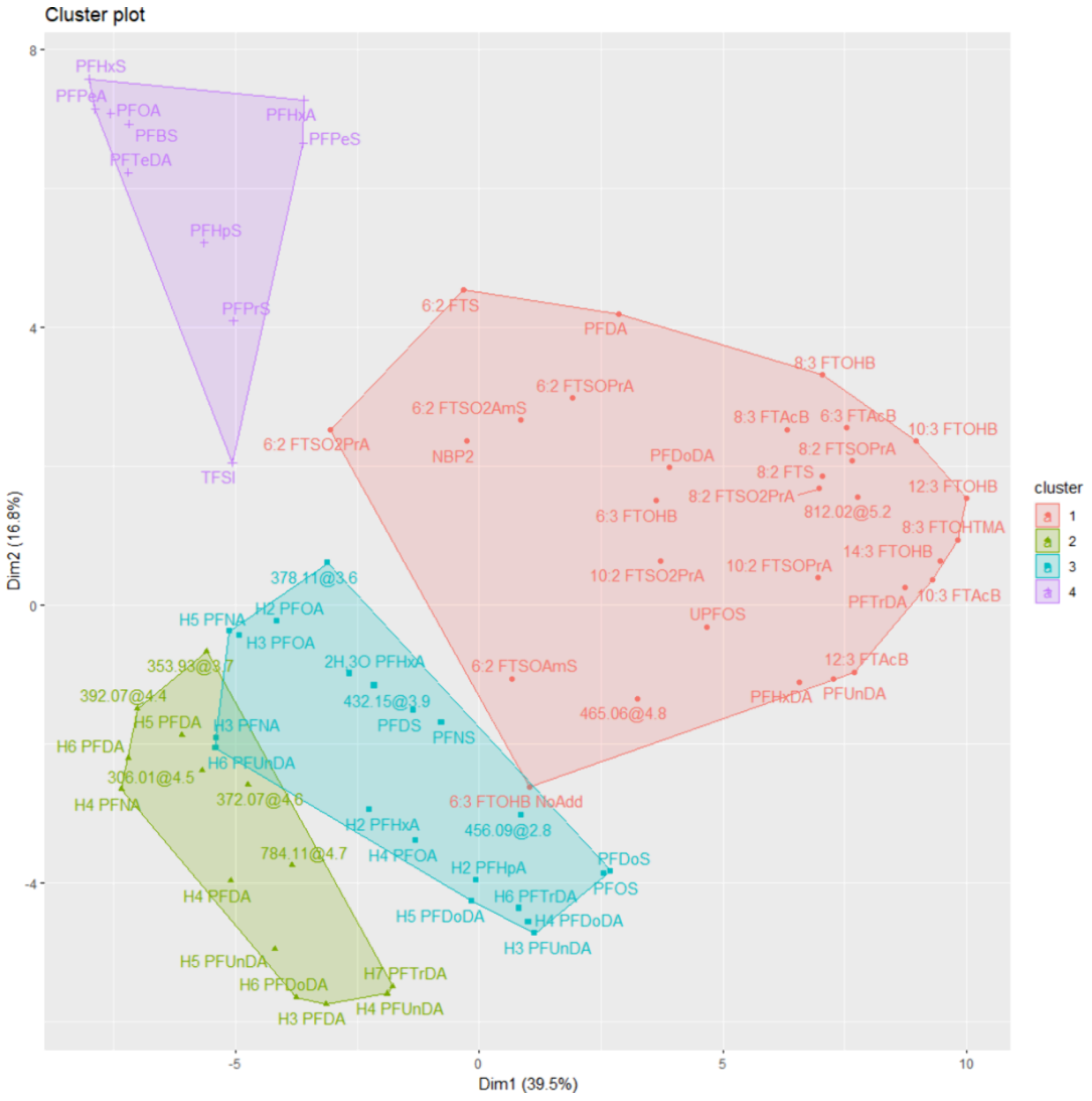


Figure 3. K-medoids cluster plot of compound groups from coabundance analysis. Colors indicate k-medoids group membership as given in Figure S21. Abbreviated names are linked to compound names in Table S4. Group membership (Mem) is determined by calculating the angular distance between t-statistics and utilizing k-medoids clustering on $n = 4$ groups. Four intuitive clusters are identified.

bistriflimide (confidence 4), although the grouping is fairly weak as t-statistics involving bistriflimide are small (Figure S21), and it is a visual outlier in the k-medoids cluster plot.

Co-Abundance Interpretation. The coabundance analysis yields three groups dominated by legacy carboxylates and sulfonates (groups 3 and 4) and/or H-substituted carboxylates (groups 2 and 3). The coabundance of these three groups could be driven either by common uses or common transport processes, with both carboxylates and sulfonates being dominantly anions at environmental pH values. Group 1 contains two substantially different functional groups: betaines, which are zwitterionic to cationic at low pH, and carboxylic acids as part of a sulfinyl or sulfonyl propanoic acid headgroup, which is anionic to neutral at low pH. Thus, their sorption and transport properties should differ substantially. The reason for their coabundance in tap water is thus more likely related to use in common sources, and/or emergence from common transformation pathways of precursors. Existence of homologues of both subclasses with only even numbers of perfluorinated carbons suggests a fluorotelomer synthesis process, as opposed to electrochemical fluorination which produces homologues of both even and odd chain length.[27,73] Other fluorotelomers (the n:2 FTS and n:2 FTRAmS subclasses) also cluster with compound group 1. The limited reported observations of the sulfinyl and sulfonyl propanoic acids point to one possible pathway to their presence in tap water: treatment/transformation of AFFF-related compounds such as the thioether 3-[2-(perfluoroalkyl)ethylthio]propionate present in DuPont Zonyl FSA,[27] or 6:2 fluorotelomer thioether amido sulfonate, a component of several AFFF formulations which can hydrolyze to give the same compound.[52,66] Reports of the betaines measured in this study are also limited, although two betaine-

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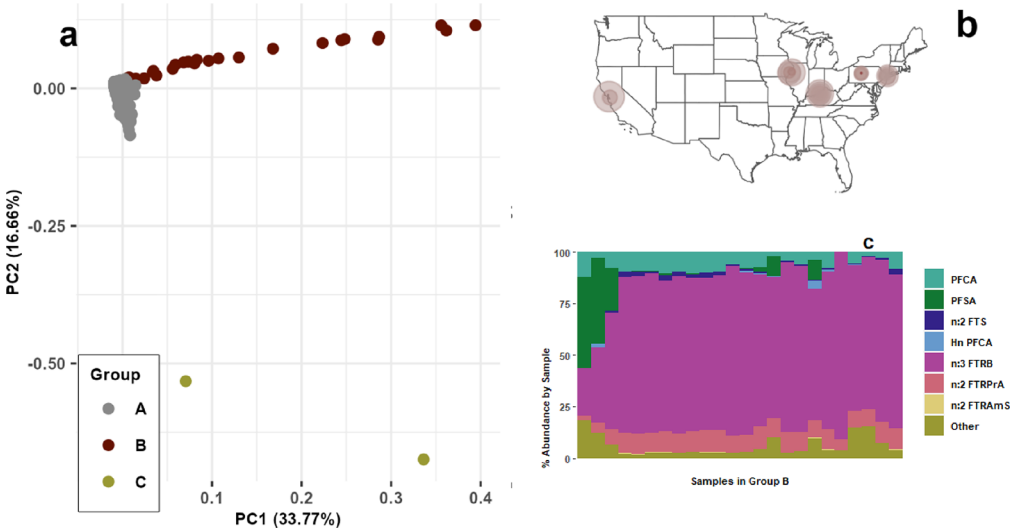


Figure 4. Sample similarity by PCA (a) and selection of samples with similar PFAS profiles (b, c). (a) PCA of log-transformed PFAS abundances in each sample. Groups were identified visually. (b) Locations and total PFAS abundances of $n = 24$ samples in group B. Color and size scales are given in Figure 2. (c) Rainbow plot of abundance by chemical subclass in group B samples, ordered by PC1.

containing homologous series (lacking R = OH, OCOCH3, and with an odd number of perfluorinated carbons) were identified as components in Buckeye AFFF, and more general zwitterionic components were found in several AFFF mixtures.[74] The betaines specific to this study have only been reported as the acetoxy members (R = OCOCH3) with n = 4, 6, 8, 10, or 12 perfluorinated carbons, in patents for betaine manufacture by DuPont,[75] water block prevention surfactant,[65] low-alcohol foam for toiletry/cosmetic/pharmaceutical application,[76] and inkjet recording sheets.[77] As the only reported use for 6:2 fluorotelomer thioether amido sulfonate is in AFFF, the most likely origin of tap water betaines and propanoic acids is AFFF, modified by oxidation and hydrolysis either in ambient conditions, at wastewater treatment plants, or in samples due to acidic preservation. This general conclusion is in agreement with spatial analyses of targeted PFAS data. For instance, PFAS detection frequency in US drinking water sampled under the Third Unregulated Contaminant Monitoring Rule (UCMR3) is significantly associated with a number of nearby potential sources, including industrial sites where PFAS are manufactured or used, military firefighting training facilities, and wastewater treatment plants whose effluent may serve as a source.[78]

In addition to common uses including industry sources e.g.,[33,78−81] processes including water treatment and distribution may influence the coabundance of PFAS in drinking water. Septic systems influence the chemical contaminants including PFAS detecting in private drinking water wells on Cape Cod, Massachusetts.[82] Public water system treatment technologies[83] and point-of-use filters[32] have a wide range of effectiveness on both legacy and emerging PFAS, including the potential to increase PFAS loads of shorter-chain PFAS by desorption, leaching, and/or transformation. Sorption, transportation, and leaching from drinking water system pipes may also contribute to PFAS in drinking water over time.[84] The extent to which each factor influences PFAS abundances in AHHS II samples cannot be determined, but the presence of emerging PFAS in samples implies those PFAS persist through the treatment process, and the lack of reported uses other than AFFF for several of the identified fluorotelomer species leaves AFFF as the most likely ultimate source.

3.4. Sample Similarity. Sample composition under principal component analysis (PCA) dimensionality reduction is shown in Figure 4a. Chemical components as weighting vectors are shown in Figure S20. Two groups of samples (A, B) and two outliers (C) emerge from this analysis. Most samples fall into sample group A, with relatively minor variation in log-transformed PFAS abundances, the majority of which is subparallel to PC2. The most relevant weightings corresponding to variation within group A are mostly PFSAs and H-substituted PFCAs (Figure S20). Sample group B consists of 24 samples, varying mostly subparallel to PC1, largely in the direction mostly of betaine and sulfinyl/sulfonyl propanoic acid weightings. This group is essentially collinear, as would be expected of end-member mixing with a source of a single composition, although dimensionality reduction may mask variability within this group. Contamination within these samples is not relegated to a single location, rather to five separate locations in this study (Figure 4b). The locations (identified by ZIP only) are a mix of urban and rural locations. Broadly, the ZIP codes are fairly close to airstrips, airports, or railyards where AFFF may have been used, but this is also true of many samples outside sample group B. Group B samples are dominated by betaines and sulfinyl/sulfonyl propanoic acids, with an increase in their proportional abundance concomitant with a decrease in PFSA abundance moving from left to right along PC1 (Figure 4c).

The composition of the most contaminated samples (>1 standard deviation above mean total PFAS abundance) is shown in Figure 5. Four samples are contaminated predominantly by legacy PFCAs and PFSAs. Ten samples are contaminated largely by betaines and sulfinyl/sulfonyl propanoic acids, with some PFCA contamination as well (but limited presence of PFSAs). Two samples are contaminated mostly by H-substituted PFCAs.

The outlier samples (Figure 4, Group C) correspond to the two samples in Figure 5 with an abundance of H-substituted PFCAs. One sample contains virtually no other PFAS, while the other contains legacy PFAS as well. From single-location detections for each of these two patterns, it is not straightforward to discern the cause of their substantially different PFAS profiles. Possibilities may include transformation of PFCAs by environmental processes, especially anoxic conditions seemingly

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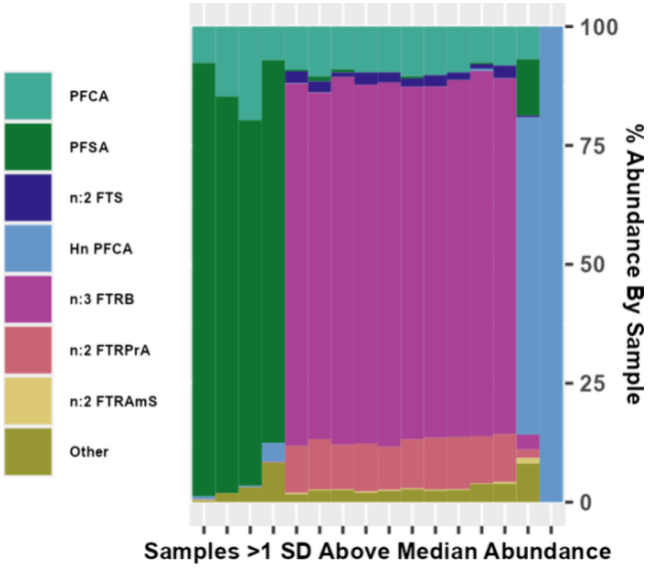


Figure 5. Stacked bar (rainbow) plot of PFAS instrument abundances in the most contaminated samples (i.e., those samples with a total abundance at least 1 standard deviation above the mean abundance, n = 16). Compounds grouped and colored by chemical subclass. Samples ordered by a weighting function to group similar compositions together (PFCAs and PFSAs weighted to the left; H-substituted PFCAs weighted to the right). Sample locations are plotted on a map of the US in Figure S22.

necessary to reduce PFCAs; or use of particular product formulations. Abundances of H-substituted PFCAs containing an even versus an odd number of H atoms are similar in both samples. The use of NTA is necessary to identify such outlier instances in broad sampling campaigns which would be missed by most targeted approaches.

3.5. Detecting PFAS in Other Environmental Samples.

Targeted methods are an important tool for monitoring and quantifying the most well-studied legacy PFAS. Application of targeted methods in sampling efforts such as UCMR5 will help make communities safer by tracking the abundance of known PFAS over time. However, targeted methods cannot detect off-target chemicals, leaving open the possibility that PFAS contamination is a much broader problem. NTA studies can help narrow the focus from the thousands of potential PFAS to only those which are present in tap water. In this study, we find 75 PFAS in a nationally representative tap water survey. Eighteen of these PFAS are routinely tracked by targeted methods. The remaining 57 are generally present in only a small number of samples, but are high in abundance in those samples.

There is evidence of contamination of a small proportion of tap water samples by emerging PFAS, dominated by a likely AFFF-derived source. However, these results do not suggest that targeted PFAS panels used in tap water screening need to be expanded by dozens of compounds. Instead, a combination of NTA and use of one or two standards as surrogates may enable screening in the context of the current tiered testing strategy. For instance, H-substituted PFCAs largely occur together, and one or two could serve as stand-in calibrants should targeted quantitation prove necessary. Moreover, the structural similarity to PFCAs probably makes their ionization behavior similar to PFCAs with the same chain length; their retention times are, as expected, observed to be similar (Table S4). Surrogate PFAS with similar functional group and perfluorinated chain length can act as surrogates for PFAS lacking analytical standards for semiquantitative analysis (e.g.,[85,86]) although associated uncertainties in concentration estimates may be large in some cases.[69,87] Properties of the sulfinyl/sulfonyl propanoic acids and betaines are likely more divergent, preventing confident semiquantitation without standards. However, their coabundance and likely common origin from transformation of AFFF mean that semiquantitation of one member of one series could, by proxy, allow semiquantitation of other members, should the correlations between their measured abundances hold in other samples and sample types. As common components of the most contaminated samples in this data set, it will be important to evaluate the presence of AFFF-related compound group 1 in additional samples, especially whether or not they appear in nonacidified samples, as it is possible some of these PFAS are transformation products. That being said, we would expect transformation by oxidation (both in drinking water treatment by chlorination, as well as in samples acidified with nitric acid) to generally decrease abundances of fluorotelomer compounds. It will continue to be important to subject tap water samples to NTA, as the distribution of emerging PFAS and emerging contaminants in general will change over time.

More work remains to be done to characterize the best way to select proxies in a statistically rigorous manner. As an example, three example correlations from compound group 1 (Figure S25) demonstrate that the most abundant or most centrally located compounds within a cluster are not necessarily representative of abundances of the rest of the group in all cases. However, when 6:3 2-hydroxy betaine abundances are high, abundances of the other compounds shown are also high. Thus, 6:3 2-hydroxy betaine could be a good proxy indicator for high levels of compound group 1 contamination. The most representative compound may depend on the desired nature of the proxy, e.g., whether an indicator of high contamination or any contamination at all is desired, or a chemical whose absence is a good indicator of the absence of contamination. A thorough examination of the nature of proxy selection would require in-depth analysis of multiple NTA studies across multiple use cases, such as case-control, impacted community, time series, or representative surveys.

Given the subclasses of PFAS present in this study, it is prudent to analyze environmental samples for PFAS monitoring in both positive and negative polarity modes whenever possible, or at least to thoroughly screen for adducts, especially where AFFF is a possible contributor to contamination. Of the ten betaines identified in this study, only one was detected directly in ESI (−) mode, and because its abundance was 2 orders of magnitude lower than the adduct pairs, it was not selected for DDA MS^2 fragmentation. Additionally, 8:3 fluorotelomer 2-hydroxy trimethylammonium, a permanent cation, was only detected as a diadduct with two anions. In both cases, identification was aided by a distinctive mass difference of 2.97 amu between the nitrate and acetate adducts in our samples, with the nitrate present only due to the addition of nitric acid (Section S1). It may be useful to add multiple volatile anions (e.g., acetate, nitrate, formate) to aid identification of zwitterionic and cationic species in negative mode if full treatment of samples in positive mode is not possible, as formation of multiple adduct pairs would give distinctive mass separations between coeluting adduct peaks. This work and others'[26,27,88,89] demonstrate the prevalence of zwitterionic and cationic PFAS in environmental samples.

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LC-ESI-MS methods can only detect and quantify species amenable to chromatographic separation in the liquid phase, and to ionization. This excludes species which are too volatile (such as fluorotelomer alcohols), difficult to protonate/deprotonate (such as perfluoroalkane sulfonamides, which in targeted methods are usually detected as adducts), or have unsuitable polarity for LC analysis and elute too early for reversed phase LC (e.g., ultrashort-chain PFAS). These biases should be remembered when using any LC-ESI-MS methods, but especially non-targeted ones, which can provide the illusion of presenting a "complete picture" of compound presence due to the number of different analytes detected. As such, there are possibly, even likely, PFAS in these samples that went undetected by our methods.

Retrospective NTA of samples collected originally for different purposes has the potential to expand our ability to track emerging contaminants. However, analysis of samples collected for different purposes also carries challenges related to the inability to fully control for some sources of contamination. We describe some of these issues in Section S4. We highlight that use of replicate analysis, batchwise method blanks, and sample order randomization should be used to rule out contamination from field collection, sample preparation, and analysis carryover.

In this study, we applied UPLC-ESI-HRMS/MS to tap water samples originally gathered during the American Healthy Homes Survey II for metals analysis. We performed a combination of suspect-screening and NTA approaches to identify PFAS in the samples, finding several subclasses of PFAS including H-substituted PFCAs, betaines, and sulfinyl/sulfonyl propanoic acids which are not captured by targeted methods. Through a combination of network and other statistical analyses, we determined coabundance and spatial patterns regarding the prevalence of these PFAS. The betaines and sulfinyl/sulfonyl propanoic acids especially contaminate a small number of localized but geographically diverse samples at a high level, with their presence likely due to transformation of AFFF. It is uncertain whether these compounds were present originally or transformed from another undetected precursor due to acidic sample preservation. QAP analysis in particular helped to elucidate patterns of coabundance and thus potential for coexposure, as well as provide evidence for PFAS contaminant origin when coupled with functional use information. However, because of the sparse nature of the functional use information in this instance, the assignment of contaminant origin is provisional. Overall, current targeted panels of PFAS do a good job identifying the most abundant PFAS in most samples, especially for samples from communities that are not highly impacted by contaminated drinking water. However, non-targeted analysis has an important potential role in monitoring for emerging or unrecognized contaminants, especially regarding diverse emerging PFAS.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.4c08037.

Metadata tables for identified PFAS and non-PFAS (XLSX)

Supplementary methods, feature identification and confidence description, spatial statistics, field and laboratory contamination discussion, annotated mass spectra, chromatography plot of PFOA analogues, feature abundance distribution plots, Spearman's rank and QAP-derived t-statistic coabundance matrices, PCA loading plot, targeted concentration/NTA abundance scatterplots, and run sequence plots (PDF)

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Notes

The authors declare no competing financial interest.

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